



UNIVERSITY OF PISA

Master of Science in Computer Engineering

PERFORMANCE AND EVALUATION OF
COMPUTER SYSTEMS AND NETWORKS

5G Connected High-Speed Train

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Chapter 1

Introduction

A high-speed **train** moves at constant **V** along a railway of length **L** = 10km. Once it reaches the end of the railway it reappears at the starting point, continuing its motion seamlessly.

The train provides WiFi connectivity to **N** passengers through an onboard wireless **Access Point (AP)**, and the **AP** is connected to a 5G radio network consisting of **M** base stations (**BSs**) placed alongside the railway at distance **D** = **L/M** between each other.

Each **passenger** generates packets of size **S** bytes every **T** seconds, and transmits them to the AP. In turn, the AP forwards packets to its closest BS. The AP can transmit only one packet at a time following a First In First Out (FIFO) policy, at a rate $\mathbf{R} = \mathbf{C} * e^{-(\mathbf{d}/1000)^2}$, where **C** is the maximum data rate (in bytes/s) and **d** is the distance between the AP and the BS.

Before starting each transmission, the AP checks which BS is the closest. If the closest BS differs from the current serving BS, the AP needs to perform a handover operation that takes **H** seconds. The AP cannot perform any transmission during the handover and resumes transmissions after the handover is complete.

Evaluate at least the overall system throughput for various values of **N** and **M** in the following scenario(s):

- Exponential distribution for both **T** and **S**.

1.1 Objectives

The main objective of our study is to analyze the performance of the system while varying some parameters such as the number of passengers on the train and the number of base stations placed all along the railway. We mainly focused our attention on studying the throughput of the system (the numbers of bytes per second transmitted) and the average response time (the time elapsed between a packet generation and the end of its transmission).

1.2 Performance Indexes

The indexes we used to track the performance of the system are:

- The overall system **Throughput**, defined as follows

$$\frac{\textit{Total Bytes Sent}}{\textit{Measurement Period}}$$

with *Measurement Period* defined as *Simulation Time – Warm Up Period*.

- The **Response time**, defined as the time elapsed between the packet generation and the end of its transmission. In other word, is calculated as the sum of the time the packet spends in the queue (waiting time) and the time required for its transmission, from the AP to the nearest BS (transmission time).

Chapter 2

Modelling

2.1 General Assumptions

To model this scenario we made the following assumptions:

1. The propagation delay between the Access Point and a Base Station is considered equal to zero
2. The transmission time is calculated using the rate at the instant the transmission starts, and remains unchanged regardless of how the rate changes afterwards (due to the distance)
3. The Access Point's queue is assumed to be infinite
4. We have chosen to use a uniform random variable to represent the handover delay

2.2 Preliminary Validation

A preliminary validation is required to ensure the model's accuracy, so the previous general assumptions must be analyzed:

1. The propagation delay, denoted as T_{prop} , represents the time required for the electromagnetic wave to propagate from the AP to the nearest BS and is calculated as the ratio between the physical distance (d) and the speed of light (c):

$$T_{prop} = \frac{\text{distance } (d)}{\text{propagation speed}}$$

Since the distance (in the order of hundreds of meters) is relatively short compared to the propagation speed ($c \approx 3 \times 10^8 \text{ m/s}$), the resulting time is negligible to the respect of the other system latency, such as the packet generation or transmission time.

2. We recall that the transmission rate can be calculated using the following formula:

$$Rate = C \times e^{-\left(\frac{d}{1000}\right)^2}$$

Although the train travels at a constant speed, causing the distance d appearing in the formula to vary, the change in distance during the transmission of a single packet is minimal. Therefore, the variation in the data rate is considered negligible, and the rate is assumed to remain constant for the duration of the transmission of a single packet.

3. We are not interested in how the system's throughput is influenced by packet loss due to the queue being full, so we can assume the queue as infinite.
4. The handover delay depends primarily on the transmission protocol and the physical channel properties. Since all Base Stations are identical and data transmission is suspended during the handover operation, the processing time is consistent across the network. Consequently, we model the handover delay as a uniform random variable with a narrow range of $0.1s$, reflecting the deterministic nature of the protocol.

2.3 Factors

- **N**: Number of passengers on the train
- **M**: Number of Base Stations placed along the railway
- **H**: Mean handover delay (s)

2.4 Parameters

- **S**: Mean size of the packets (*bytes*)
- **T**: Mean generation time of the packets (s)
- **C**: Maximum transmission rate of the AP ($\frac{\text{bytes}}{s}$)
- **V**: Train speed ($\frac{km}{h}$)
- **L**: The railway length is fixed at 10 km based on the system description

2.5 Observation

To gain a deeper insight into the system's behavior, we performed a detailed analysis of specific factors. Let C denote the maximum transmission rate (the choice of this parameter is discussed in Section 5.1). The transmission rate profile is defined as follows:

$$\text{Rate} = C \times e^{-\left(\frac{d}{1000}\right)^2}$$

This behavior is graphically illustrated in Figure 2.1.

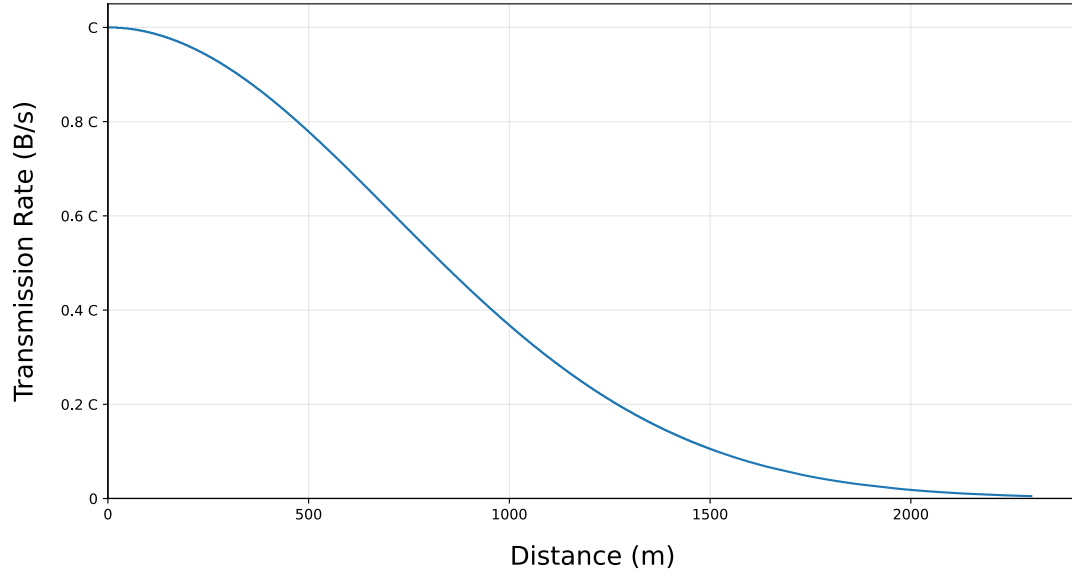


Figure 2.1: Transmission Rate Function

However, d is not a simple variable, rather it is a function of the relative distance between the AP and the nearest Base Station (BS). Considering a scenario with 5 BSs spaced 2 km apart, the maximum distance to the nearest BS is half this distance, i.e., 1 km. The behavior of d is shown in Figure 2.2.

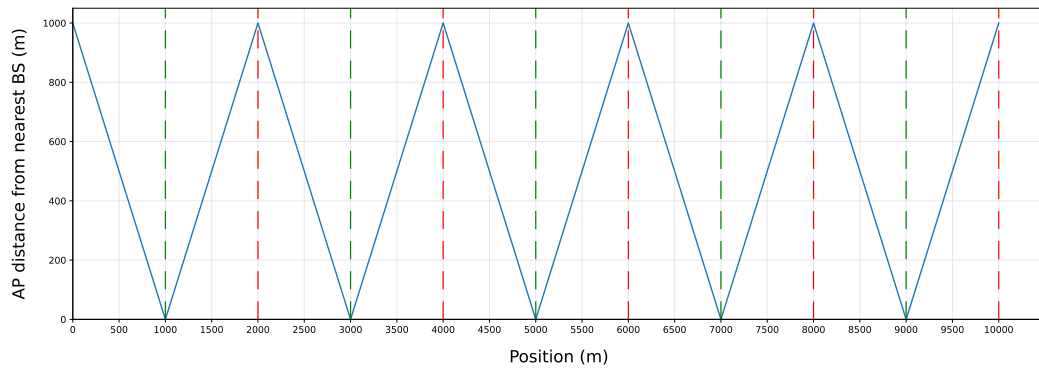


Figure 2.2: Distance between AP and nearest BS (with 5 BSs)

Given the behavior of d , we can derive the transmission rate profile. Recalling the formula, the rate exhibits an exponential trend characterized by two extremes:

- **Maximum Rate:** This is reached when the relative distance between the AP and the nearest BS is 0 m. In this case, the exponential term equals 1, and the rate equals C (highlighted by a dashed green line).
- **Minimum Rate:** This occurs when the distance is at its maximum (1 km). Here, the exponential term becomes $e^{-1} \approx 0.368$, resulting in a rate of $0.368 \times C$ (highlighted by a dashed red line).

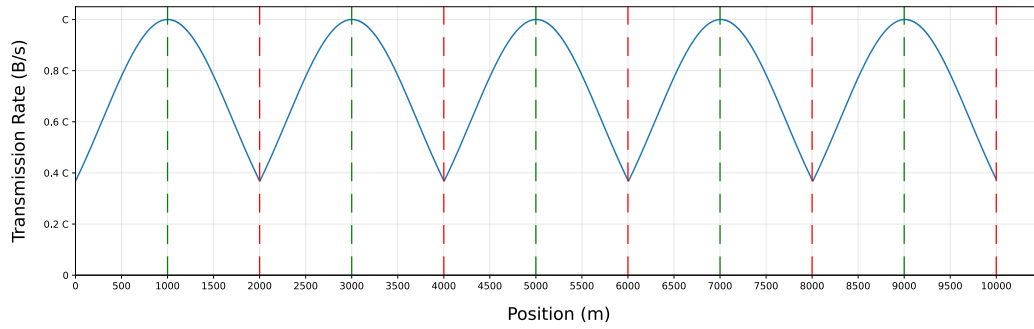


Figure 2.3: Transmission Rate (with 5 BSs)

Chapter 3

Implementation

The system has been implemented in Omnet++ by defining new modules, which reside inside the **Railway** network.

3.1 Modules

- **Base Station:** simple module which models one of the entry points of the 5G radio network. All the base stations are evenly spaced along the railway
- **Mapper:** simple module which models an omniscient entity that knows all the base stations' positions
- **Train:** compound module which models the moving vehicle. This module contains:
 - **Passenger:** simple module which models one of the N passengers
 - **Access Point:** simple module which acts as a bridge between the passengers and the base station by receiving, storing and forwarding packets
 - **Simple Mobility:** simple module which updates the train's position
 - **Debug Module:** simple module which prints out values during the graphical simulation for debug purposes. It has no effect on the simulation

3.1.1 Modules' behavior

Base Station

A base station resides along the railway and, when the train is close enough, it begins receiving the passengers' packets, forwarded by the Access Point. When the train is close to a base station, all the other base stations remain idle.

Mapper

The mapper knows and manages all the base stations' positions. At the beginning of the simulation, it calculates all the BSs' positions, and saves the index of the BS which is currently connected to the train's AP. It's a passive module, as it remains idle after initializing, unless explicitly queried.

Passenger

Every passenger is modeled as a simple packet generator. After the initialization, it periodically extracts a random waiting interval from an exponential distribution with mean T ; creates and sends a packet with a random size (extracted from an exponential distribution with mean S) using the `cPacket` message class.

Access Point

The Access Point's role is to store all the packets sent by the passengers, and then forward them one after the other to the closest BS taking $\frac{Packet\ Size}{Current\ Rate}$ seconds. It implements an infinite FIFO queue and is work conserving. At the beginning of every transmission, it directly accesses the Mapper's methods (no messages are exchanged) in order to check which BS is currently the closest. If a different BS is closer than the current one, the module extracts an handover interval from an uniform distribution (with mean H and range $0.1\ s$) and remains idle until the handover operation is completed. After that, it reverts to its standard behavior.

Simple Mobility

This module stores and updates the train's position every `trainUpdateInterval` seconds.

Chapter 4

Verification

We analyzed the implementation's behavior in order to validate and verify its correctness. We performed the following tests:

- Deterministic Test
- Degeneracy Tests
- Consistency Test
- Continuity Test

4.1 Deterministic Test

This test's objective is to evaluate the behavior of the system in a deterministic scenario. This means that there is no randomness involved, and we can check the results analytically.

In order to turn the simulation into a deterministic one, we have introduced the **train.randomness** parameter which can be set to false to run this simulation. In this way, we will prevent the code from extracting random values for the packets' size, the generation time and the handover delay. Instead, the mean value of each distribution is used as a constant. The resulting simulation parameters are defined as in Table 5.2.

The system was simulated for 1000 s using the previous data, resulting in a measured throughput of 999 bytes/s. This result is consistent with the following calculation: 10 passengers each generate a 100-byte packet every second, for a total rate of 1000 bytes/s. Although packet generation is scheduled at integer seconds from $t = 1$ s to $t = 1000$ s, the packets at $t = 1000$ s coincide with the simulation end and are not sent. Consequently, only 999,000 bytes are transmitted and, as expected, the final throughput results in 999 bytes/s.

Parameter	Value
Number of Passengers (N)	10
Number of Base Stations (M)	10
Handover Delay (H)	0 s
Train speed (V)	300 Km/h
Packet Size (S)	100 B
Generation Time (T)	1 s
Max Transmission Rate (C)	3.5 MB/s

Table 4.1: Simulation Parameters - Deterministic Test

4.2 Degeneracy Tests

These tests aim to verify the behavior of the system in extreme conditions, by degenerating some parameters, in order to ensure the system still works as expected. The following tests have been performed:

- $\mathbf{M} = \mathbf{0}$ (No Base Stations): since no base station is present, the access point can't forward any packet.
- $\mathbf{V} = \mathbf{0}$ (No Train speed): when the speed is 0 *km/h*, the train remains stationary at the starting position (0 *m*). While the access point continues to transmit, the throughput remains constant at its minimum value, since the train is located at the maximum distance from the first base station.
- $\mathbf{N} = \mathbf{0}$ (No Passengers): if there are no passengers, the AP does not receive any packet, meaning that the total throughput will be 0 *bytes/s*.
- $\mathbf{C} = \mathbf{0}$ (Max Rate = 0): if the maximum rate is 0 *bytes/s*, no packets are forwarded, resulting in a throughput of 0 *bytes/s*.
- $\mathbf{H} = \mathbf{0}$ (No Handover): if H is set to 0 *s*, no handover delay occurs when the AP switches between base stations.

By running the corresponding simulations, we verified that each test produced the expected results.

4.3 Consistency Test

The objective of this test is to validate the consistency of the system's behavior. It was performed by comparing the throughput of various simulations:

- A simulation with 20 passengers and a mean packet generation time of 0.15 *s*
- A simulation with 40 passengers (twice the number of the previous case) and a mean packet generation time of 0.3 *s* (corresponding to half the generation rate)

	Simulation A	Simulation B
Passengers	20	40
Mean Generation	0.15 <i>s</i>	0.3 <i>s</i>
Throughput	53045 B/s	53038 B/s

Table 4.2: First Consistency Test

The throughput of the two simulations is almost identical, as expected. The two results are not exactly equal due to the randomness involved in the packet generation.

If we compare the following runs instead:

- A simulation with 40 passengers, and a mean packet generation time of 0.3 *s*
- A simulation with 40 passengers, and a mean packet generation time of 0.15 *s* (corresponding to twice the generation rate)
- A simulation with 20 passengers (half the number of the previous cases), and a mean packet generation time of 0.3 *s*

	Simulation B	Simulation C	Simulation D
Passengers	40	40	20
Mean Generation	0.3 <i>s</i>	0.15 <i>s</i>	0.3 <i>s</i>
Throughput	53038 B/s	106980 B/s	27148 B/s

Table 4.3: Second Consistency Test

we notice that the throughput almost doubles in the second run and is halved in the third one, as expected.

4.4 Continuity Test

This test verifies if a slight variation of the input translates into a slight variation of the output. The following parameters were used in our simulation:

Parameter	Value
Number of Passengers (N)	20
Number of Base Stations (M)	15
Mean Handover Delay (H)	0.5 s
Train speed (V)	300 Km/h
Mean Packet Size (S)	{300 B, 325 B, 350 B}
Mean Generation Time (T)	0.03 s
Max Transmission Rate (C)	3.5 MB/s

Table 4.4: Simulation Parameters - Continuity Test

The results obtained are the following:

	Simulation A	Simulation B	Simulation C
Mean Packet Size	300 B	325 B	350 B
Throughput	199502 B/s	216155 B/s	232808 B/s

Table 4.5: Continuity Test

Chapter 5

Calibration

The calibration phase represents a fundamental aspect of system modeling. An appropriate selection of factors' range allows us to simulate realistic scenarios and evaluate the system's sensitivity to parameter variations. Furthermore, correctly defining the warm-up period and simulation time ensures that the analysis reflects the system's behavior in a steady-state condition.

5.1 Factors calibration

Mean Packet size (S) The Maximum Transmission Unit (MTU) specified by the IEEE 802.3 Ethernet protocol is 1500 *bytes*. Modeling the packet size as an exponential random variable (as requested) with a mean of 400 bytes, we calculate the probability of generating a packet exceeding the MTU as follows:

$$P\{X \geq 1500\} = e^{-\frac{1500}{400}} = 0.023$$

A probability of 2.30% is sufficiently low to consider $S = 400$ bytes an appropriate choice for the mean packet size.

Mean packet generation time (T) We selected an interval of 30 *ms* to realistically model passenger traffic, as this matches the standard transmission rates of common real-time applications like video streaming (30 fps) and voice calls.

Train speed (V) High-speed trains typically operate at speeds ranging from 250 to 350 km/h. In our model, the train moves at a constant speed V fixed at 300 km/h. This value represents an average of the typical operating range, ensuring a realistic simulation scenario.

Number of passengers (N) The analysis focuses on various scenarios characterized by varying passenger loads. Excluding the zero-passenger case, which was previously addressed in the degeneracy test, we examined configurations ranging

from low occupancy (50 passengers) up to a near-train-capacity scenario (400 passengers).

Number of base stations (M) The system was analyzed under different configurations; in particular, the number of Base Stations (BSs) varied between 4 and 16. Consequently, the maximum distance between the Access Point (AP) and the nearest BS ranged from 1250 m down to 312.5 m.

Increasing the number of base stations has a double effect on the system: it increases the handover frequency, since the train changes base stations more often, and increases the average output rate, since the interval in which the data rate oscillates is shorter.

Maximum Transmission Rate (C) To determine an appropriate value for C , we estimated the link capacity. By fixing the maximum transmission rate at 3.5 MB/s and assuming a mean packet size of 400 B, the system at its best (the distance from a base station is zero) can transmit approximately:

$$\frac{\text{Max Transmission Rate (C)}}{\text{Mean Packet Size (S)}} = \frac{3.5 \text{ MB/s}}{400 \text{ B}} = 8750 \text{ packets/s}$$

This capacity is appropriate, as it exceeds the aggregate traffic load. Indeed, assuming a scenario with 200 passengers and a mean packet generation time of 0.03 s, the generated traffic is:

$$\frac{\text{Number of Passengers (N)}}{\text{Mean Generation Time (T)}} = \frac{200}{0.03} \approx 6667 \text{ packets/s}$$

These theoretical estimates will not perfectly coincide with the actual system throughput, as we are currently neglecting the impact of handover delays and the distance from which depends the rate variation (factors that will be the center of our analysis).

Furthermore, considering the passenger count ranges from 50 to 400, the system load varies significantly:

- Best Case (50 passengers): The generated traffic (≈ 1667 packets/s) is well below the capacity, so the system is expected to be stable.
- Worst Case (400 passengers): The traffic ($\approx 13\,333$ packets/s) exceeds the limit, leading to an unstable scenario (the queue and response time grow to infinity).

We intentionally included this overload scenario to fully analyze the system's behavior under stress; in this case, we expect the throughput to be bounded solely by the maximum output rate.

Mean Handover Delay (H) To analyze the system's sensitivity to handover delay¹, we considered three representative mean values:

- Best case: 0.3 s
- Average case: 0.7 s
- Worst case: 1.1 s

Based on the parameters chosen earlier, we can estimate the number of packets that accumulate in the queue during the handover operation, when transmission is temporarily suspended. Assuming $N = 200$, the results obtained by applying the following formula are reported in Table 5.1:

$$N_{add_packets} = \frac{Handover\ Mean\ (H)}{Mean\ Generation\ Time\ (T)} \times Number\ of\ Passengers\ (N)$$

Mean Handover Delay	0.3	0.7	1.1
Additional packets queue	2000	≈ 4667	≈ 7333

Table 5.1: Estimated Packet Accumulation vs. Mean Handover Delay

Considering the maximum transmission rate of 8750 packets/s (Paragraph 5.1), the estimated recovery times are 0.23s, 0.53s, and 0.84s, respectively. These values represent realistic and manageable durations for the system to recover from the handover operation.

All parameters/factors and their corresponding values are summarized in the following table:

Parameter	Value
Number of Passengers (N)	{50, 100, 200, 300, 400}
Number of Base Stations (M)	{4, 7, 10, 13, 16}
Mean Handover Delay (H)	{0.3, 0.7, 1.1} s
Train speed (V)	300 Km/h
Mean Packet Size (S)	400 B
Mean Generation Time (T)	0.03 s
Max Transmission Rate (C)	3.5 MB/s

Table 5.2: Factors calibration

¹We have already discussed in section 2.2 the reason for modeling the handover delay as a uniform random variable

5.2 Warmup Period

To determine the appropriate warm-up period, we generated two separate plots, one for each performance index we intended to analyze. We set the simulation time to 1500 s (as detailed in the following section) and fixed the number of repetitions at 15.

The data shown in Figure 5.1 was derived by computing the mean of throughput samples, recorded every 0.01 s.

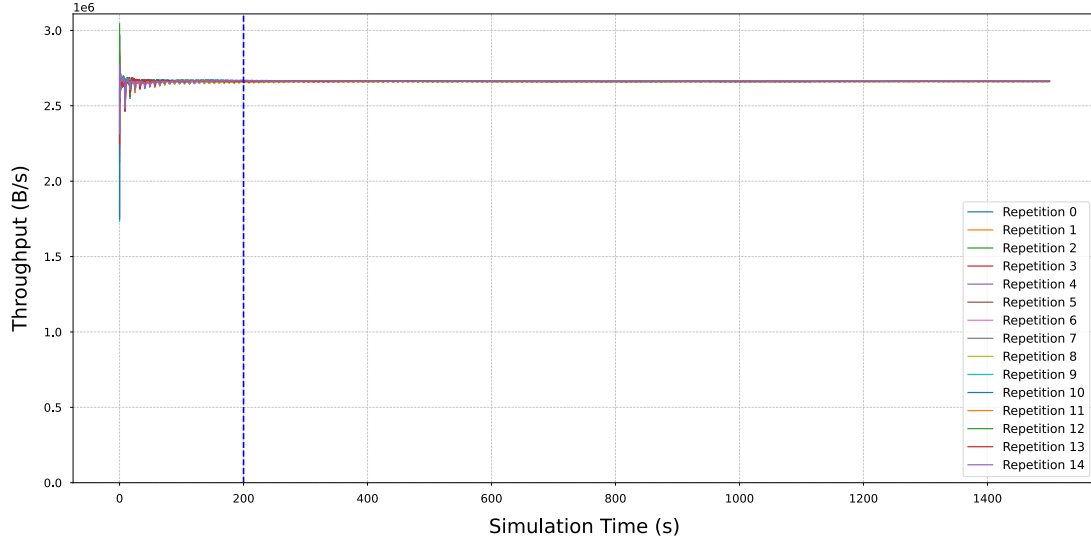


Figure 5.1: Mean Throughput plot

It is important to note that the values displayed on the ordinate axes (y-axes) in Figure 5.1 must be multiplied by a factor of 10^6 , as indicated in the upper-left corner of the plot.

As far as the second performance index (response time) is concerned, Figure 5.2 shows the mean response time of the system.

Analyzing the response time plot is particularly revealing: it indicates that an appropriate warm-up period, after which the system behavior stabilizes, is $T = 400$ s (highlighted by a dashed red line). Although the throughput stabilizes earlier (e.g., at $t = 200$ s), a single warm-up value is required for the simulation. Consequently, we selected the maximum of the two to ensure stability across all metrics:

$$T_{warmup} = \max\{200s, 400s\} = 400s$$

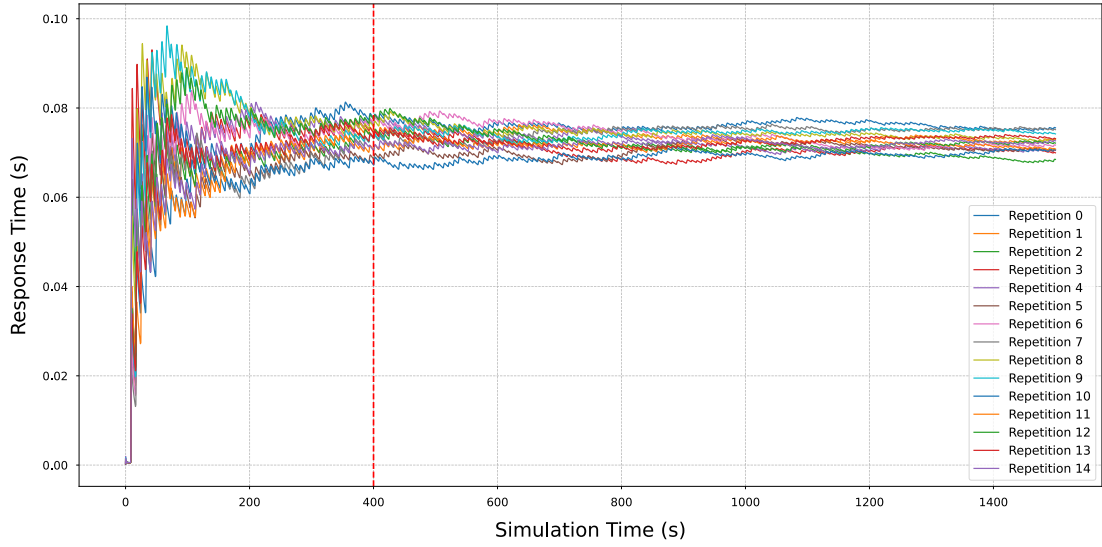


Figure 5.2: Mean Response Time Plot

5.2.1 Simulation Time

To determine an appropriate simulation time, we selected a total time of 1800 s with a warm-up period of 400 s. By applying these parameters to the limiting scenarios of 50 and 400 passengers, the average number of generated packets can be estimated using the following formula:

$$estimated_packets = \frac{sim_time - warmup_period}{Mean\ Generation\ Delay} \times N.Passengers$$

where N is the number of passengers on the train and T is the mean packet generation time. This yields approximately 2.3 million and 18.6 million packets, respectively. In the average case of 200 passengers, about 9.3 million packets are generated.

Given this simulation time, and a fixed train speed, we can obtain how many handovers will occur in the simulation with the following calculation:

$$interval_H = \frac{10km}{N.Base\ Stations \times Speed}, \quad num_H = \frac{sim_time}{interval_H}$$

In all three scenarios, this volume of data provides a sufficient statistical sample, allowing for valid and reliable conclusions for each configuration.

Chapter 6

Experiments Design

The experiments were conducted to assess the system's behavior in terms of throughput and response time, by varying the number of passengers (N), the mean handover delay (H), and the number of base stations (M) along the railway. The parameters were varied according to the values reported in Section 5.1, which are summarized in Table 6.1.

Parameter	Values
Number of Passengers (N)	{50, 100, 200, 300, 400}
Number of Base Stations (M)	{4, 7, 10, 13, 16}
Mean Handover Delay (H)	{0.3, 0.7, 1.1} s

Table 6.1: Simulation Parameter Ranges

6.1 Confidence Intervals

To ensure the correctness of the results, confidence intervals were computed using the formula:

$$[\bar{X} - \frac{Std_{dev}}{\sqrt{R}} \cdot Z_{\frac{a}{2}} ; \bar{X} + \frac{Std_{dev}}{\sqrt{R}} \cdot Z_{\frac{a}{2}}]$$

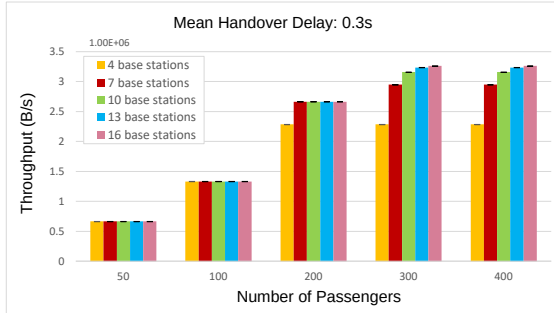
where Std_{dev} is the sample standard deviation, $R = 30$ is the number of repetitions and $(1 - a) = 99.9\%$ the confidence level chosen.

6.2 Throughput Analysis

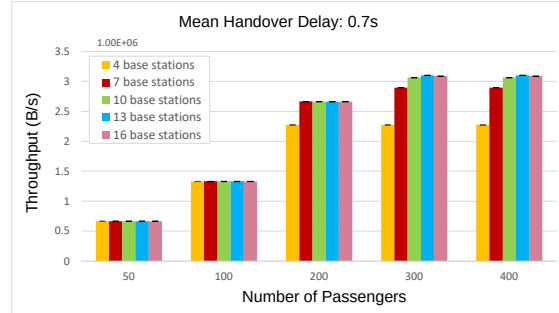
Given that we considered 5 values for N , 5 for M and 3 for H , resulting in a total of $5 \times 5 \times 3 = 75$ combinations, we performed a full factorial analysis. Each simulation was executed with 30 repetitions (total: 2250 runs) to ensure a complete and

meaningful analysis of the system throughput.

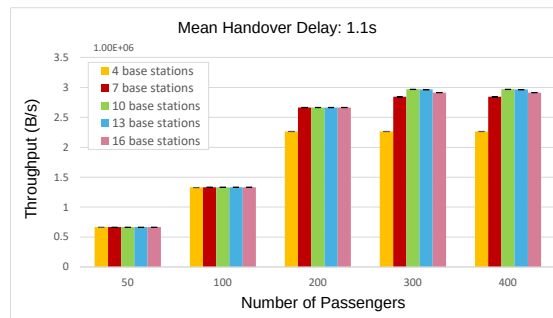
With the mean handover delay fixed, we plotted the throughput as a function of the number of passengers on the train, where each column of the bar chart represents a different number of base stations.



(a) Throughput bar chart
Mean Handover Delay: 0.3s



(b) Throughput bar chart
Mean Handover Delay: 0.7s



(c) Throughput bar chart
Mean Handover Delay: 1.1s

Low Load

The first thing that can be noticed is that for a small number of passengers (50 and 100), the throughput is roughly the same, regardless of the number of base stations and the mean handover delay. This is due to the fact that the output rate is higher than the input rate, therefore the system is stable and the throughput depends solely on the input rate, which in turn depends on the number of passengers.

Medium Load

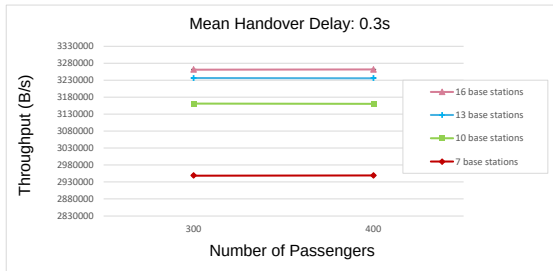
As the number of passengers increases to 200, the number of base stations starts to impact the throughput. This is due to the fact that the input rates become higher than the output rates when we have 4 BSs, while it remains smaller with an higher number of BSs. In this second case, the system is stable and adding more base stations does nothing as the throughput depends, as in the low load cases, on the number of passengers only.

High Load

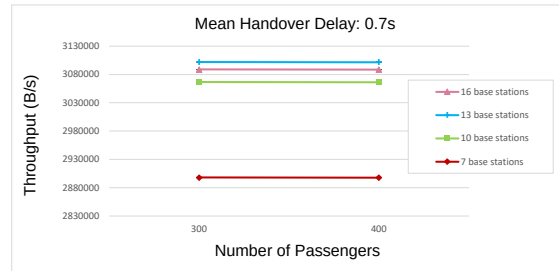
With 300 and 400 passengers, the system reaches its limit in all configurations, therefore the throughput becomes independent of the input rate. At this point increasing the number of base stations is the factor that matters most, as it increases the average output rate. Here we can see an interesting aspect of the system: increasing the number of base stations has diminishing returns, as the average output rate does not increase in a linear fashion.

The handover frequency also plays a big role, in fact more base stations equals more time spent waiting for handover operations to complete. While increasing the number of BSs, at a certain point the output rate peaks, but eventually declines. This behavior results from a trade-off: on one hand, the number of handovers increases linearly with the number of BSs stations. On the other hand, the distance between the AP and the nearest BSs decreases, pushing the minimum rate closer to the maximum rate C . However, the gain in rate gets smaller, so the handover delay ends up having the biggest impact.

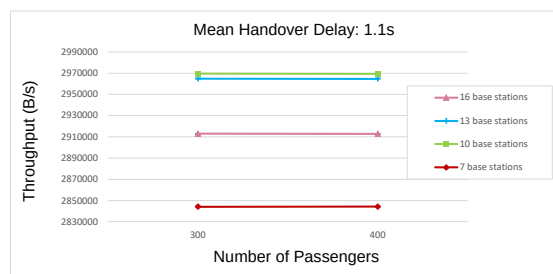
This is most noticeable when the handover time is medium-high, as less base stations are required to achieve this effect. Increasing the handover lowers and shifts the peak to the left, showing once again that the best base station configuration is strongly linked to the handover duration. This can be seen clearly in the graphs 6.2a, 6.2b and 6.2c where we get a closer look at the system's throughput under high loads (with a number of passengers between 300 and 400).



(a) Throughput High Load Scenario
Mean Handover Delay: 0.3s



(b) Throughput High Load Scenario
Mean Handover Delay: 0.7s



(c) Throughput High Load Scenario
Mean Handover Delay: 1.1s

As mentioned earlier, higher handover delays result in an optimal configuration with fewer BSs. We created a reduced Table 6.2 to highlight this trend.

	<i>7 base stations</i>	<i>10 base stations</i>	<i>13 base stations</i>	<i>16 base stations</i>
$H_{mean} : 0.3s$	$\approx 2.95 \text{ MB/s}$	$\approx 3.16 \text{ MB/s}$	$\approx 3.24 \text{ MB/s}$	$\approx 3.26 \text{ MB/s}$
$H_{mean} : 0.7s$	$\approx 2.28 \text{ MB/s}$	$\approx 3.07 \text{ MB/s}$	$\approx 3.10 \text{ MB/s}$	$\approx 3.09 \text{ MB/s}$
$H_{mean} : 1.1s$	$\approx 2.84 \text{ MB/s}$	$\approx 2.97 \text{ MB/s}$	$\approx 2.96 \text{ MB/s}$	$\approx 2.91 \text{ MB/s}$

Table 6.2: Throughput in the High Load scenario with 300 Passengers

Before moving to the next analysis, we want to point out that the y-axis scale varies in the three cases (Plot 6.2a, 6.2b and 6.2c). We did not use a unified scale because it would have made it difficult to separate the lines and see the differences.

6.3 Response Time Analysis

The results from the 300 and 400 passenger configurations were excluded from the average response time analysis. The reason behind this choice is simple: when the system is in a configuration with a large number of passengers, the output rate can't keep up with the input rate, resulting in the queue growing indefinitely. This means that the average response time does not stabilize and grows to infinity in a linear fashion as well: as a result, the final value depends heavily on the duration of the simulation.

In conclusion, we considered only 3 possible values for the number of passengers ($N = \{50, 100, 200\}$), instead of 5 as in the throughput case, which results in 45 configurations in total ($5 \times 3 \times 3$). With 30 repetitions for each configuration, we performed a total of 1350 runs.

50 Passengers

When the number of passengers is small, the response time is minimal. This is due to the fact that the queue is almost always empty, and heavy queuing occurs only during handover operations. As shown in Figure 6.3, there is a minimal advantage in increasing the number of base stations from 4 to 7 since the gain in output rate is higher than the delay introduced by having more frequent handovers. However, past this point, the introduced delay is higher than the rate gain, and the response time starts increasing with the number of base stations, with a steeper increase with higher handover durations.

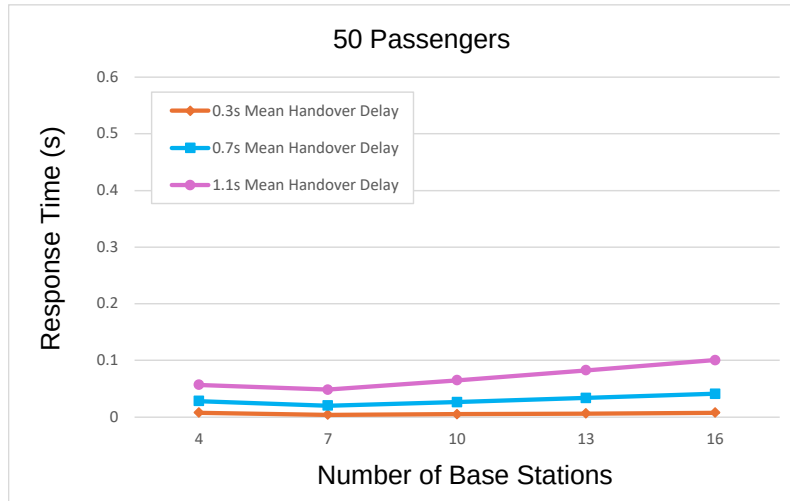


Figure 6.3: Response Time Plot with 50 Passengers Scenario

100 Passengers

With 100 passengers the input rate doubles, and the overall response time is increased. In the configuration with 4 base stations, the output rate can't keep up with the new load, therefore queuing arises even outside of handovers. Increasing the number of base stations from 4 to 7 leads to a significant reduction in response time, consistent with the trend in the previous scenario (6.3). However, the drop is much more substantial here, as heavy queuing is again solely restricted to handover intervals. Once again, past this point increasing the number of base stations results in higher response times.

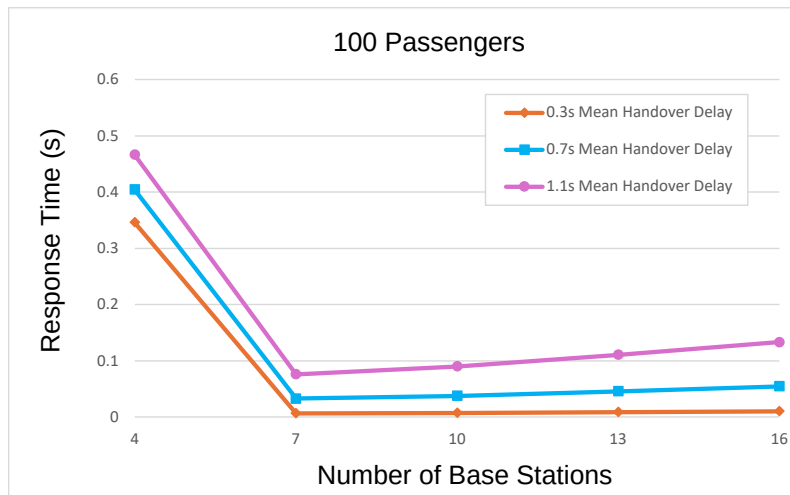


Figure 6.4: Response Time Plot with 100 Passengers Scenario

200 Passengers

In the 200-passenger scenario, the 4-base station configuration becomes unstable. Consequently, as we can see in Figure 6.5, the response time grows indefinitely, making the measured mean value heavily dependent on the simulation duration. As the number of base stations increases, the system becomes stable again. In fact, with seven base stations, the response time drops significantly and stabilizes (becoming finite again).

In the graph 6.5 the diminishing returns observed during the previous throughput analysis can be clearly noticed: increasing the number of base stations yields progressively smaller reductions in response time until a minimum is reached. After that point, as the latency introduced by handovers impacts most than the efficiency gains in the output rate. Consistent with the previous analysis, the minimum response time increases and shifts to the left as the mean handover delay grows.

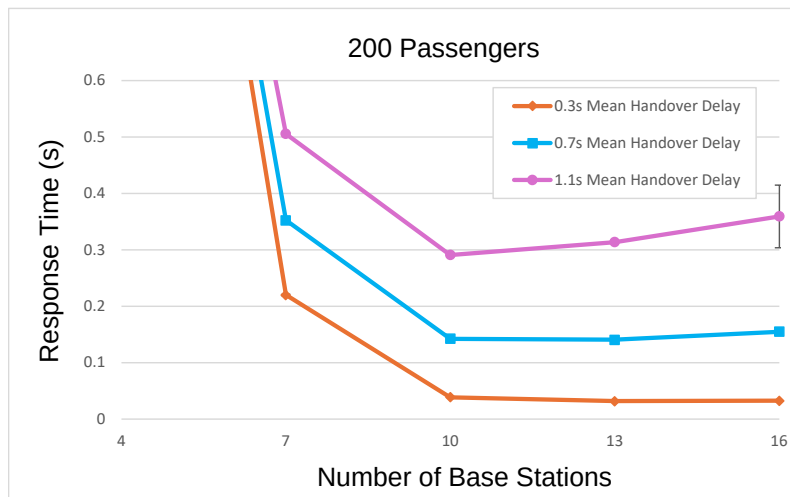


Figure 6.5: Response Time Plot with 200 Passengers Scenario

6.4 Bell Analysis

In this section¹ we visually demonstrate how the system behaves when the number of base stations is pushed to extremes. Of course, this analysis has a primary theoretical scope: a configuration of 112 BSs, where the distance between two BSs is 89m, is not a realistic case. However, it is very useful in order to understand the behavior of the system at the limit.

In fact, through this analysis we were able to show the aforementioned effect caused by the excessive increase of base stations and how it is influenced by varying mean

¹The section title refers to the shape of the graphs, not to any formal method

handover delays. To achieve these results, we evaluated the various system configurations detailed in Table 6.3.

Parameter	Value
Number of Passengers (N)	250
Number of Base Stations (M)	From 4 to 112 with step of 3
Mean Handover Delay (H)	$\{0.3, 0.7, 1.1\}$ s

Table 6.3: Factors Bell Analysis

Throughput

To evaluate throughput, a saturated (unstable) configuration is required to observe the system's maximum output rate. As we can see in graph 6.6, throughput initially grows with diminishing returns, peaks at a specific configuration and then decreases linearly. The value of the mean handover delay heavily impacts this behavior: the longer the handover duration, the steeper the decline and the optimal point (the one associated with the throughput peak) is shifted to the left.

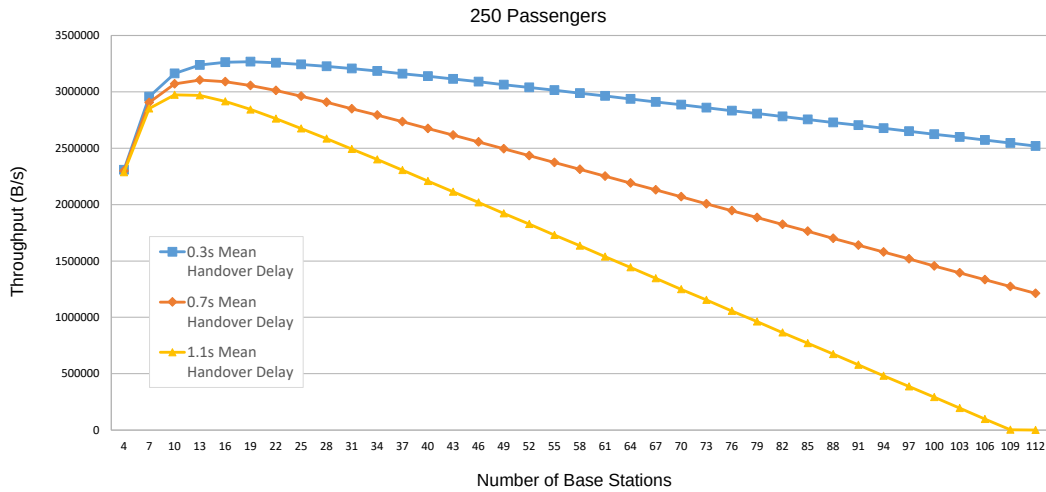


Figure 6.6: Throughput Bell Analysis

From a theoretical point of view, this behavior aligns perfectly with expectations: as the number of base stations increases, the cumulative time consumed by handover operations continues growing. Initially, as we have already deeply discussed in Section 6.2, a larger number of base stations, improves throughput: this happens because the reduced distance between the AP and the nearest BS boosts the minimum data rate. However, after the peak, handover delays begin to outweigh these benefits. Eventually, the system reaches a saturation point (such as the 1.1s

handover scenario with 112 BSs) where the train enters a continuous cycle: as soon as one handover operation finishes, the next begins, leaving no time for actual data transmission. This behavior, evident in Figure 6.6, can be confirmed analytically.

If we focus our attention on the system configuration with mean handover delay of 1.1 s and 112 BSs, we can calculate the time required for the train to travel from one BS to the next:

$$d_{AP-BS} = \frac{\text{Railway Length } (L)}{\text{Number of BS } (M)} \approx 89.286 \text{ m}$$

$$t_{BS_i-BS_{i+1}} = \frac{d_{AP-BS}}{\text{Train speed}} \approx 1.07 \text{ s}$$

We find that the train takes approximately 1.07 s to travel between two consecutive base stations, which is smaller than the time needed to perform the handover operation (1.1 s).

Response Time

As we can see in Figure 6.7, the response time exhibits a behavior that mirrors the throughput, showing a clear inverse trend. An interesting thing to notice is that the throughput function's highest values and the response time's lowest values are achieved in the same configurations. In addition, the limit scenario observed in the throughput analysis is evident here as well: when there is no data transmission (as in the configuration with 112 BSs and a 1.1s mean handover delay), the response time diverges to infinity.

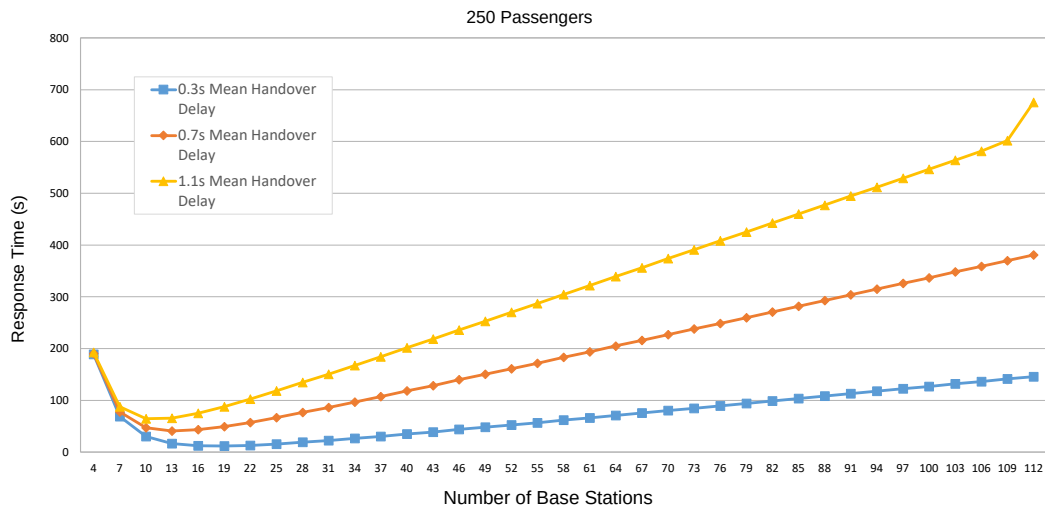


Figure 6.7: Response Time Bell Analysis

Chapter 7

Conclusions

This study provided an analysis of 5G Connected High-Speed Train, highlighting the critical need to balance infrastructure sizing (the number of base stations) with the delay introduced by handover operations. The experiments revealed a dual behavior of the system:

- **Low Load (50-100 passengers):** the system remains stable, and throughput depends almost exclusively on the input rate, which is a function of the number of passengers (N). In this scenario, increasing the number of base stations (M) has a negligible effect on performance.
- **High Load (300-400 passengers):** The system reaches saturation, with throughput limited by the maximum transmission capacity, which depends on the number of base stations (M) and on the value of the mean handover delay. In fact, increasing the number of base stations improves performance only up to an optimal point, after which diminishing returns are observed. In addition, as the mean handover delay (H) increases, the peak performance shifts towards configurations with fewer base stations.

In conclusion, the simulations confirm that there is no single absolute optimal configuration. Maximizing throughput and minimizing response times requires a trade-off between the number of BSs (that ensure a high data rate) and handover frequency (which results in periods of inactivity).